I2200: Digital Image processing

Lecture 6: Image Restoration

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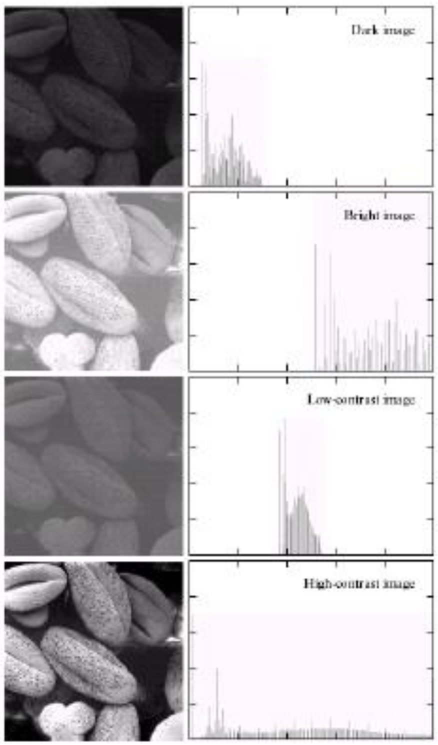
Thanks to G&W website and Lexing Xie for slide materials

We have covered …

Image sensing



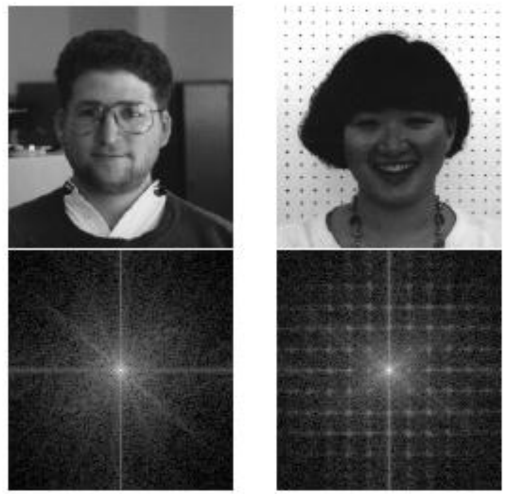


Image Transform and Filtering

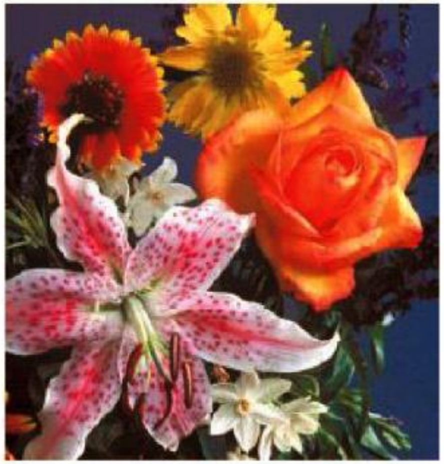
Spatial Domain processing

Image Restoration

Outline

- What is image restoration
  - Scope, history and applications
  - A model for (linear) image degradation
- Restoration from noise
  - Different types of noise
  - Examples of restoration operations
- Restoration from linear degradation
  - Inverse and pseudo-inverse filtering
  - Wiener filters
  - Blind de-convolution
- Geometric distortion and its corrections

Degraded images



Original image

Blurred image

- What caused the image to blur?
- Can we improve the image, or "undo" the effects?

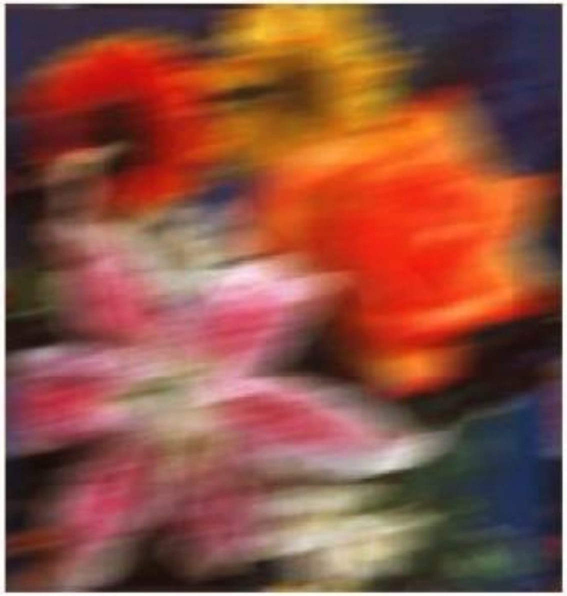
Image Restoration

- Image restoration is to "compensate for" or "undo" defects which degrade an image.
- Degradations:
  - motion blur
  - noise
  - camera defocus
  - distortion
  - …

http://www.owlnet.rice.edu/~elec539/Projects99/BACH/proj2/intro.html

Image Restoration VS Enhancement

Original image



Blurred image

- Image enhancement: "improve" an image subjectively.
- Image restoration: remove distortion from image in order to go back to the "original" → objective process.

Image restoration

- started from the 1950s
- application domains
  - Scientific explorations
  - Legal investigations
  - Film making and archival
  - Image and video (de-)coding
  - …
  - Consumer photography

- related problem: image reconstruction in radio astronomy, radar imaging and tomography

[Banham and Katsaggelos 97]

A model for image distortion

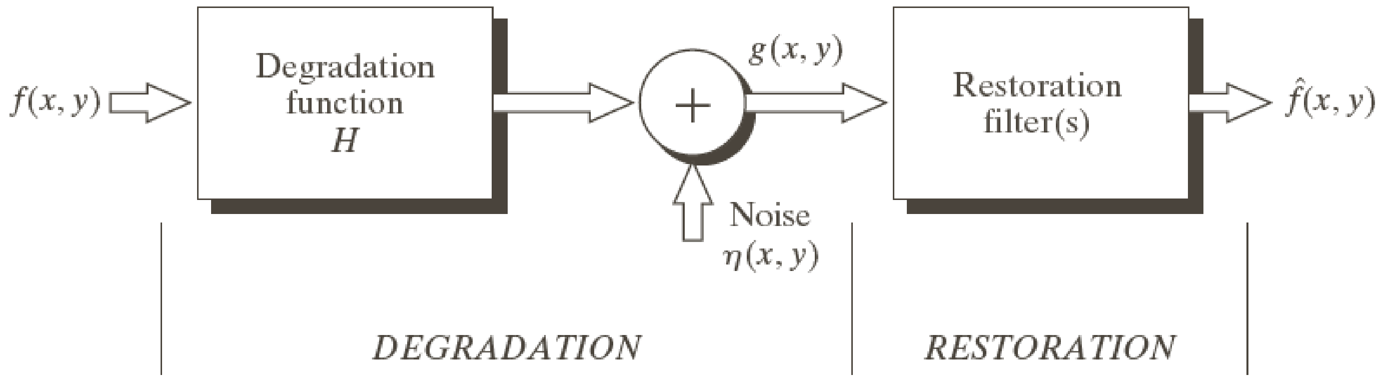


FIGURE 5.1 A model of the image degradation/restoration process.

$$g(x,y) = H[f(x,y)] + \eta(x,y)$$

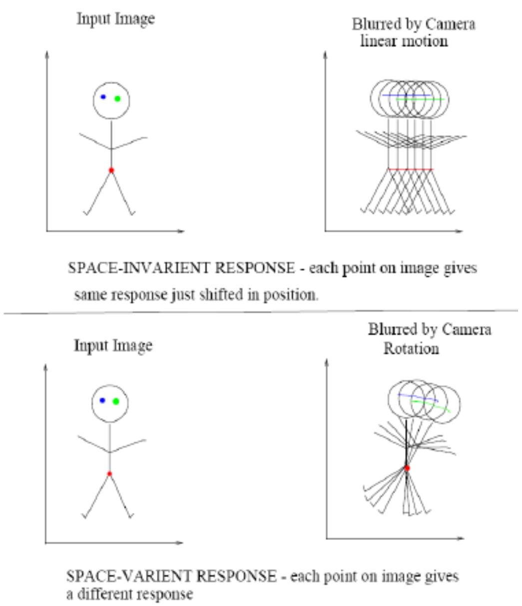
$\rightarrow$ design restoration filters such that $\hat{f}(x,y)$ is as close to $f(x,y)$ as possible.

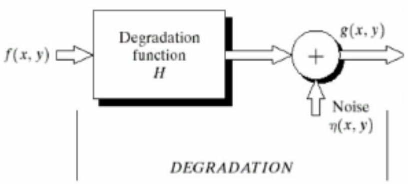
A model for image distortion

- Image restoration
  - Use a priori knowledge of the degradation
  - Modeling the degradation and apply the inverse process
  - Formulate and evaluate objective criteria of goodness

Usual assumptions for the distortion model

- Noise
  - Independent of spatial location
    - Exception: periodic noise …
  - Uncorrelated with image
- Degradation function H
  - Linear
  - Position-invariant

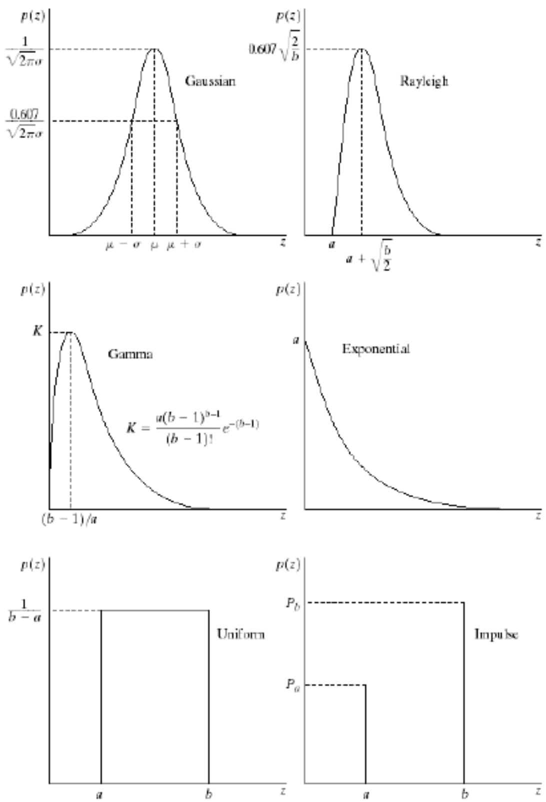




$$g(x,y) = \underbrace{H}_{=1}[f(x,y)] + \eta(x,y)$$

divide-and-conquer step #1: image degraded only by noise.

Common noise models



a b
c d
e f

FIGURE 5.2 Some important probability density functions.

Gaussian

$$p(z)=\frac{1}{\sqrt{2\pi}\sigma}e^{-(z-\mu)^2/2\sigma^2}$$

Rayleigh

$$p(z)=\frac{2}{b}(z-a)e^{-(z-a)^2/b},\ for \qquad z\geq a$$

Erlang, Gamma(a, b)

$$p(z)=\frac{a^b z^{b-1}}{(b-a)!}e^{-az},\ for \qquad z\geq 0$$

Exponential

$$p(z)=ae^{-az},\ for \qquad z\geq 0$$

→ additive noise

Salt-and-Pepper:

$$p(z) = P_a\delta(z-a) + P_b\delta(z-b)$$

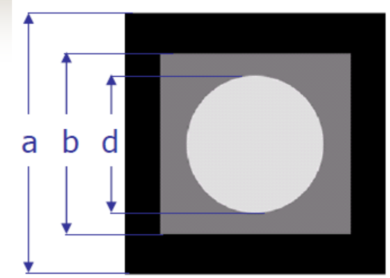
Speckle noise: $a = a_R + ja_I$

$$|g(x,y)|^2 \simeq |f(x,y)|^2|a(x,y)|^2 + \eta(x,y)$$

a_R,a_I zero mean, independent Gaussian

→ multiplicative noise on signal magnitude

The visual effects of noise



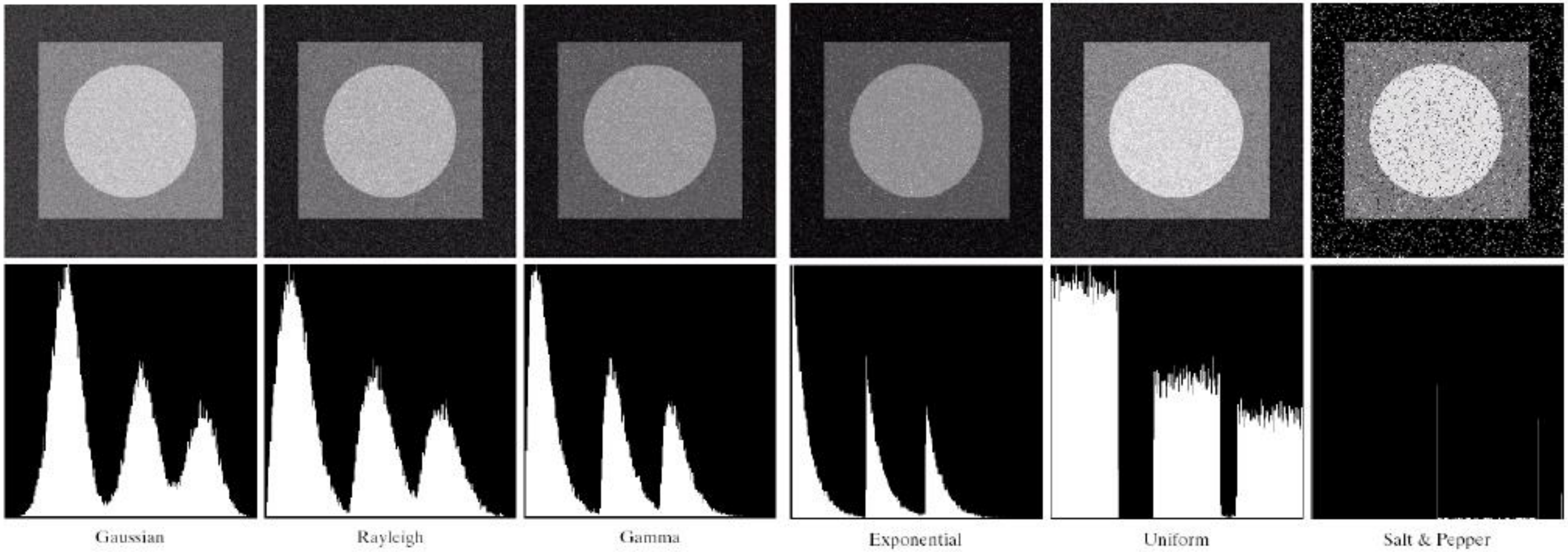


FIGURE 5.4 Images and histograms resulting from adding Gaussian, Rayleigh, and gamma noise to the image in Fig. 5.3.

FIGURE 5.4 *(Continued)* Images and histograms resulting from adding exponential, uniform, and impulse noise to the image in Fig. 5.3.

Recovering from noise

- overall process
  Observe and estimate noise type and parameters → apply optimal (spatial) filtering (if known) → observe result, adjust filter type/parameters …

- Example noise-reduction filters [G&W 5.3]
  - Mean/median filter family
  - Adaptive filter family
  - Other filter family
    - e.g. Homomorphic filtering for multiplicative noise [G&W 4.5, Jain 8.13]

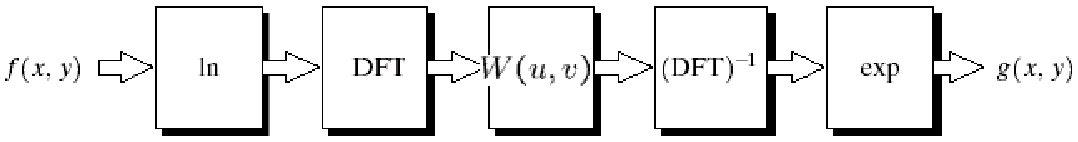
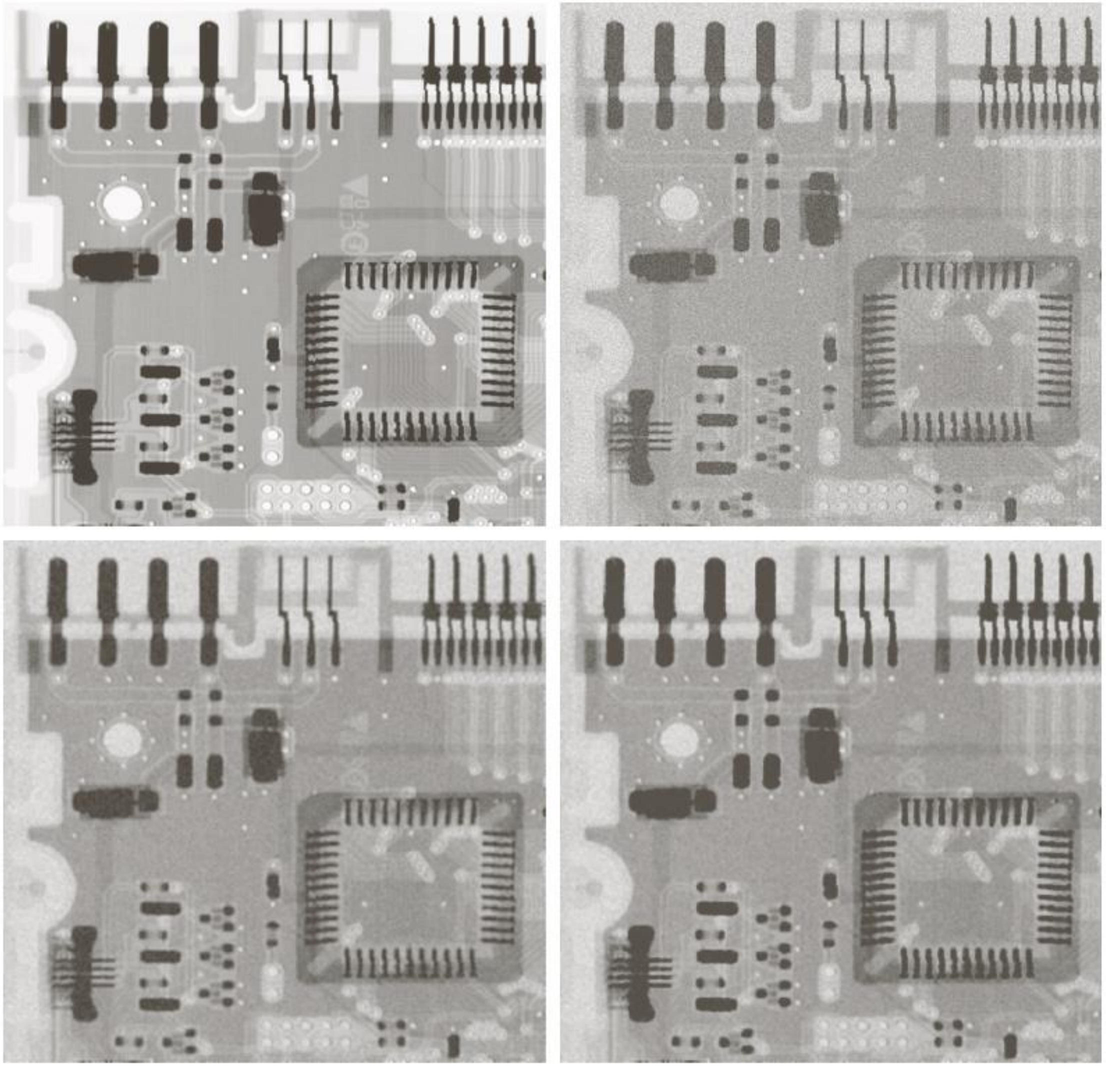


FIGURE 4.31 Homomorphic filtering approach

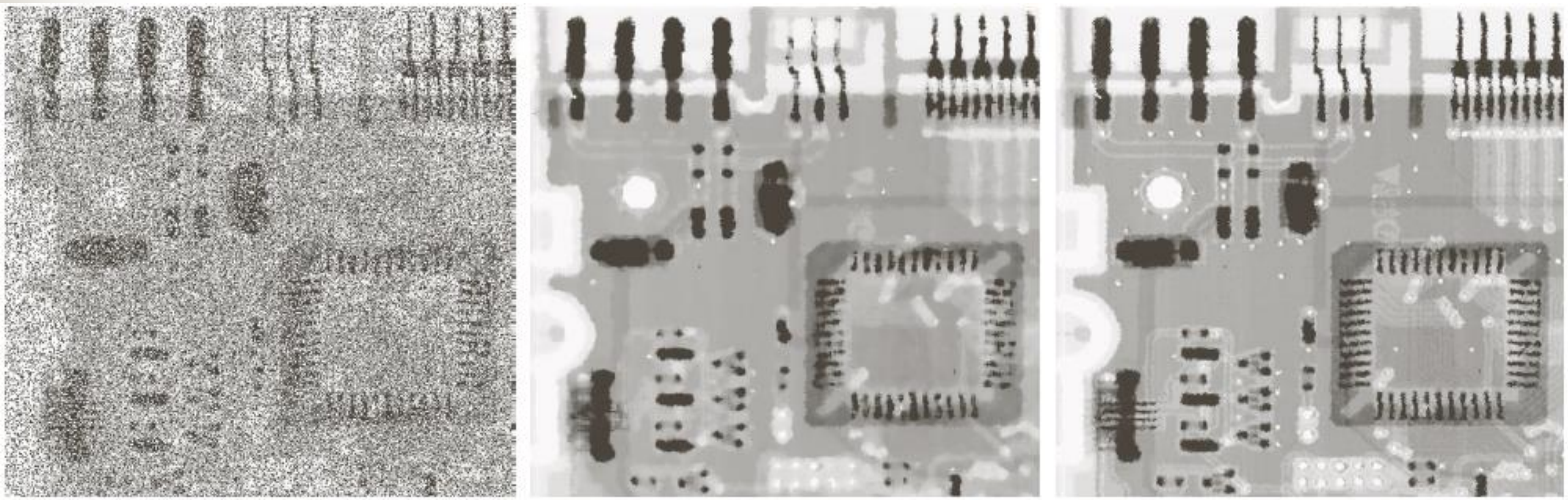
Example: Gaussian noise & mean filter



a b
c d

FIGURE 5.7 (a) X-ray image. (b) Image corrupted by additive Gaussian noise. (c) Result of filtering with an arithmetic mean filter of size 3×3. (d) Result of filtering with a geometric mean filter of the same size. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

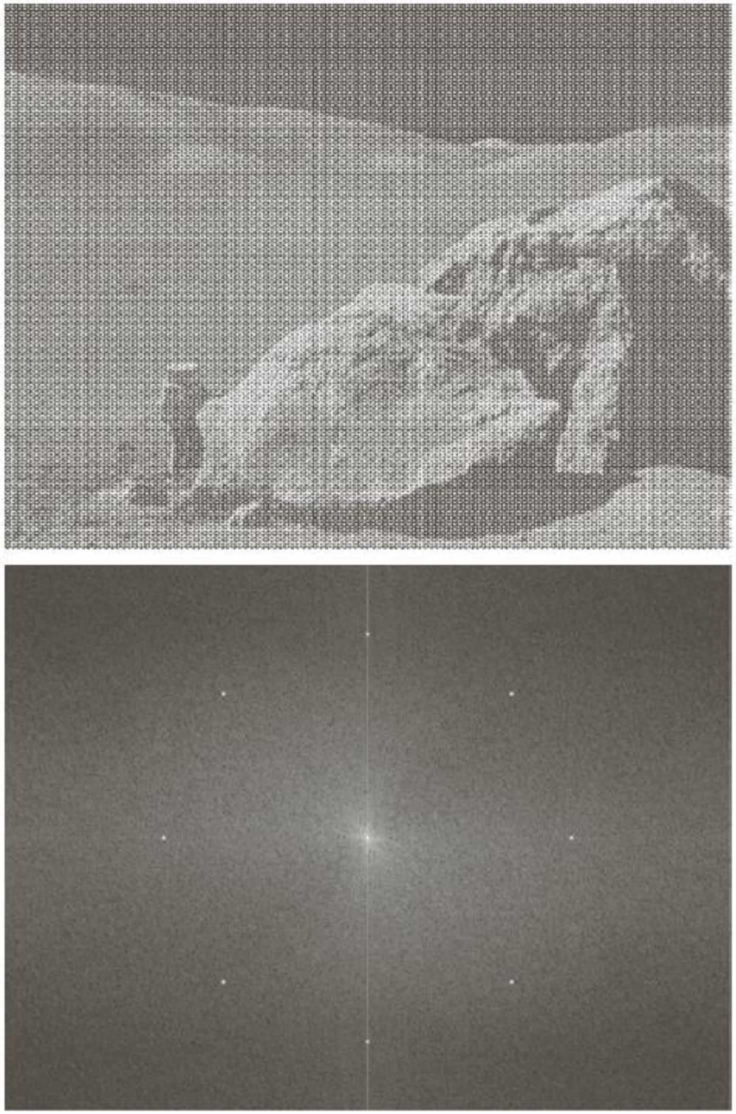
Example: salt-and-pepper noise & median filter



a b c

FIGURE 5.14 (a) Image corrupted by salt-and-pepper noise with probabilities $P_a = P_b = 0.25$. (b) Result of filtering with a 7×7 median filter. (c) Result of adaptive median filtering with $S_{max} = 7$.

Recovering from Periodic Noise



a
b

FIGURE 5.5 (a) Image corrupted by sinusoidal noise. (b) Spectrum (each pair of conjugate impulses corresponds to one sine wave). (Original image courtesy of NASA.)

Recovering from Periodic Noise in Frequency domain

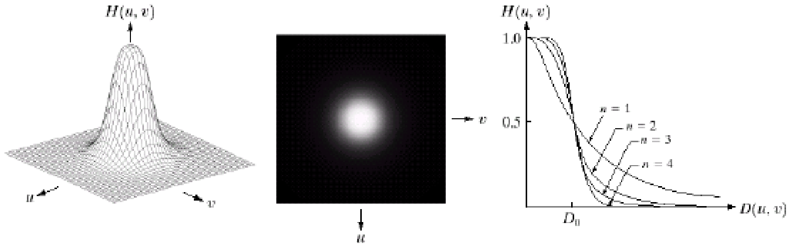
[G&W 5.4]

Recall: Butterworth LPF

$$H(u,v) = \frac{1}{1+[D(u,v)/D_0]^{2n}}$$

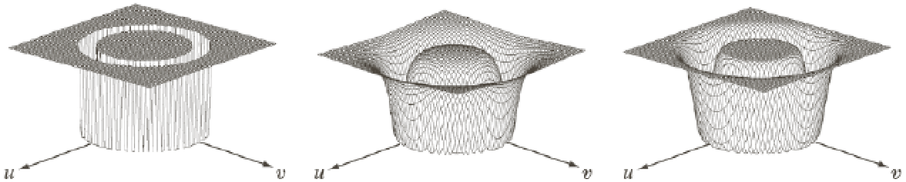
Butterworth bandreject filter

$$H(u,v) = \frac{1}{1+[\frac{D(u,v)W}{D^2(u,v)-D_0^2}]^{2n}}$$



a b c

FIGURE 4.14 (a) Perspective plot of a Butterworth lowpass filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.



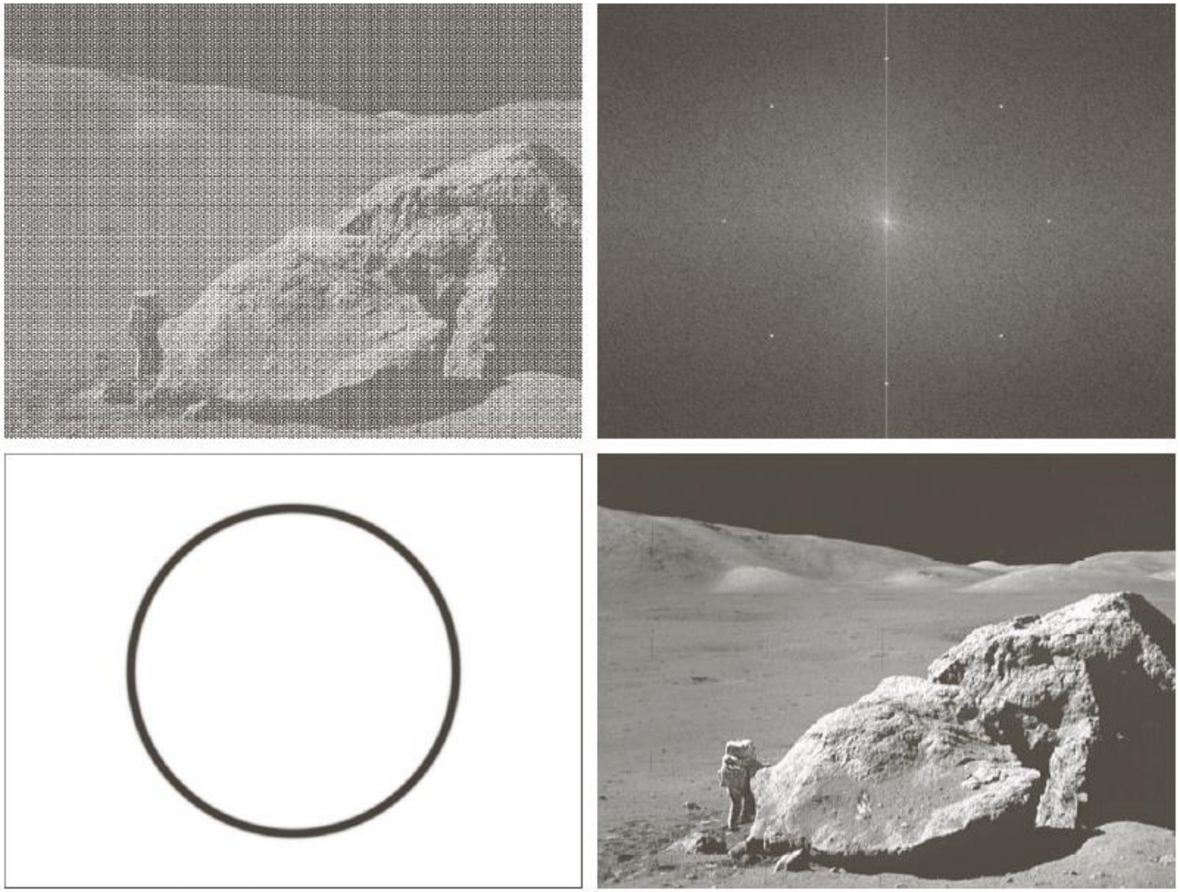
a b c

FIGURE 5.15 From left to right, perspective plots of ideal, Butterworth (of order 1), and Gaussian bandreject filters.

Example of bandreject filter

a b
c d

FIGURE 5.16 (a) Image corrupted by sinusoidal noise. (b) Spectrum of (a). (c) Butterworth bandreject filter (white represents 1). (d) Result of filtering. (Original image courtesy of NASA.)



Example of bandpass filter

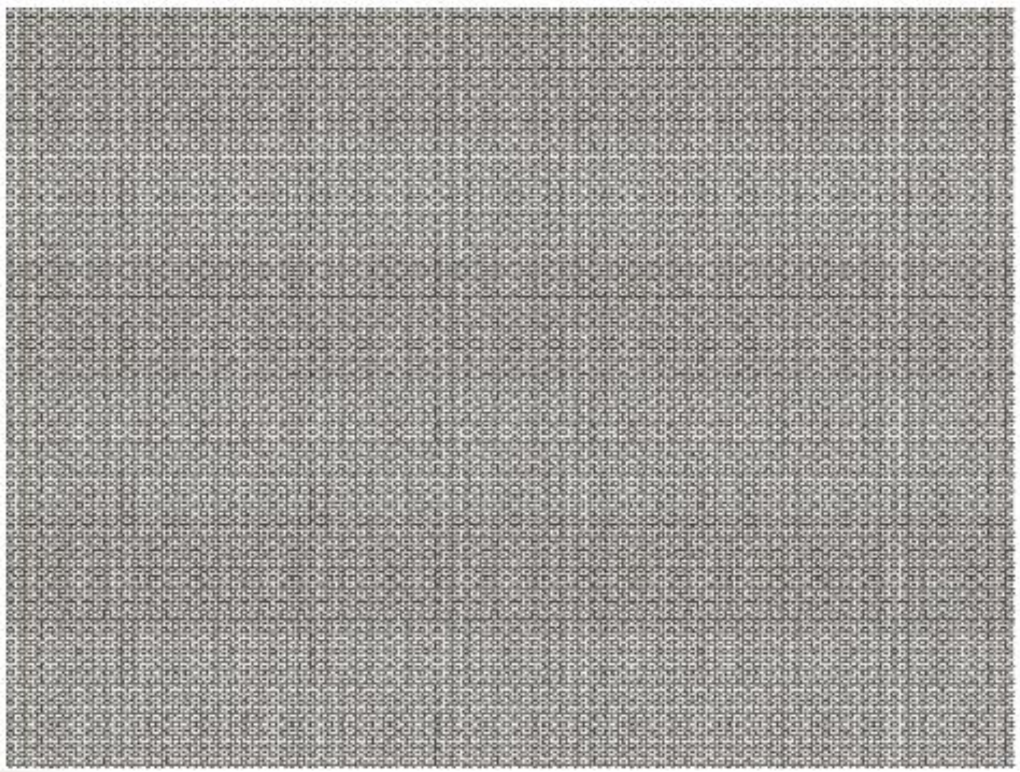


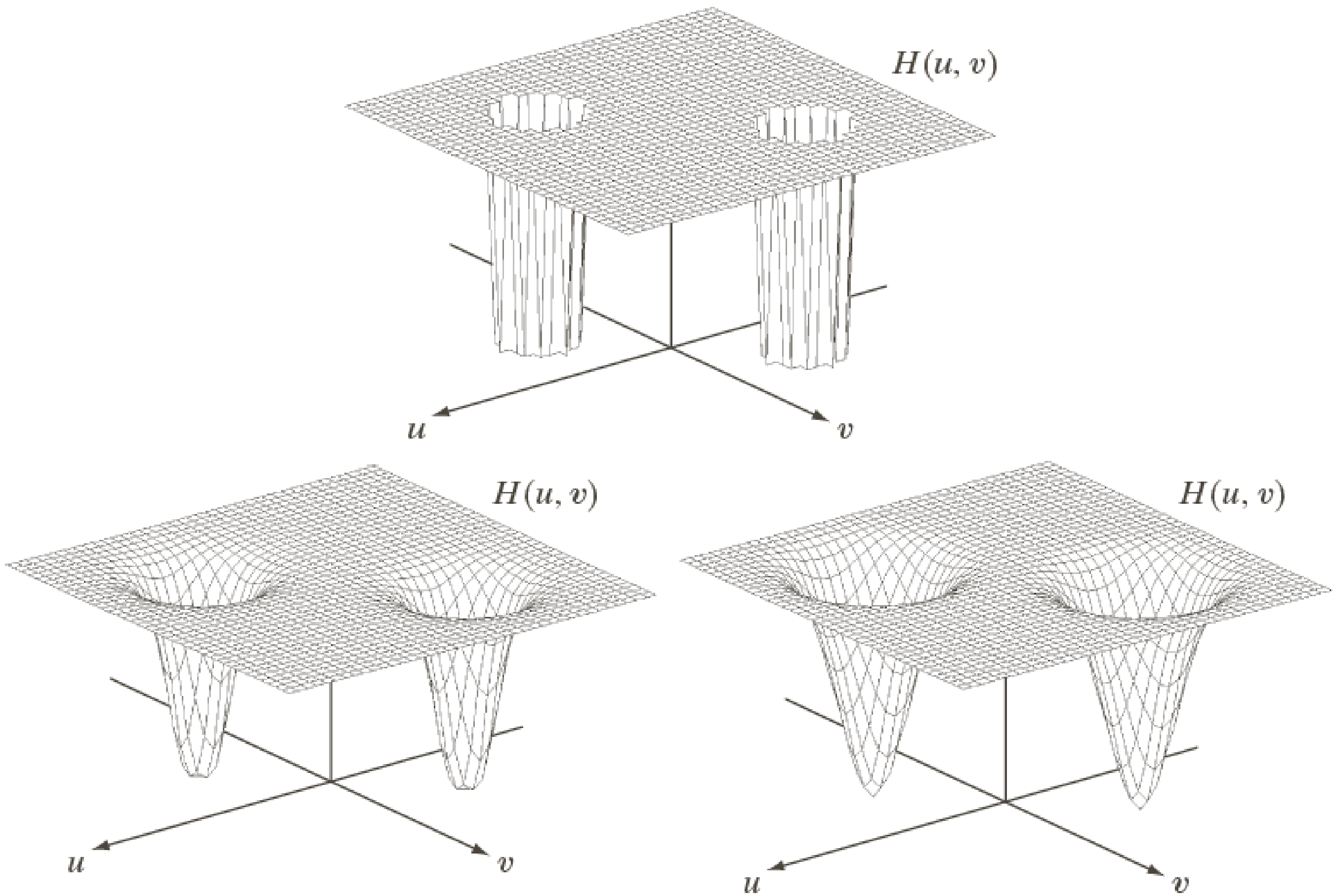
FIGURE 5.17 Noise pattern of the image in Fig. 5.16(a) obtained by bandpass filtering.

$$Hbp(u, v) = 1 - Hbr(u, v)$$

Notch filter

a
b c

FIGURE 5.18 Perspective plots of (a) ideal, (b) Butterworth (of order 2), and (c) Gaussian notch (reject) filters.



Example of notch filter

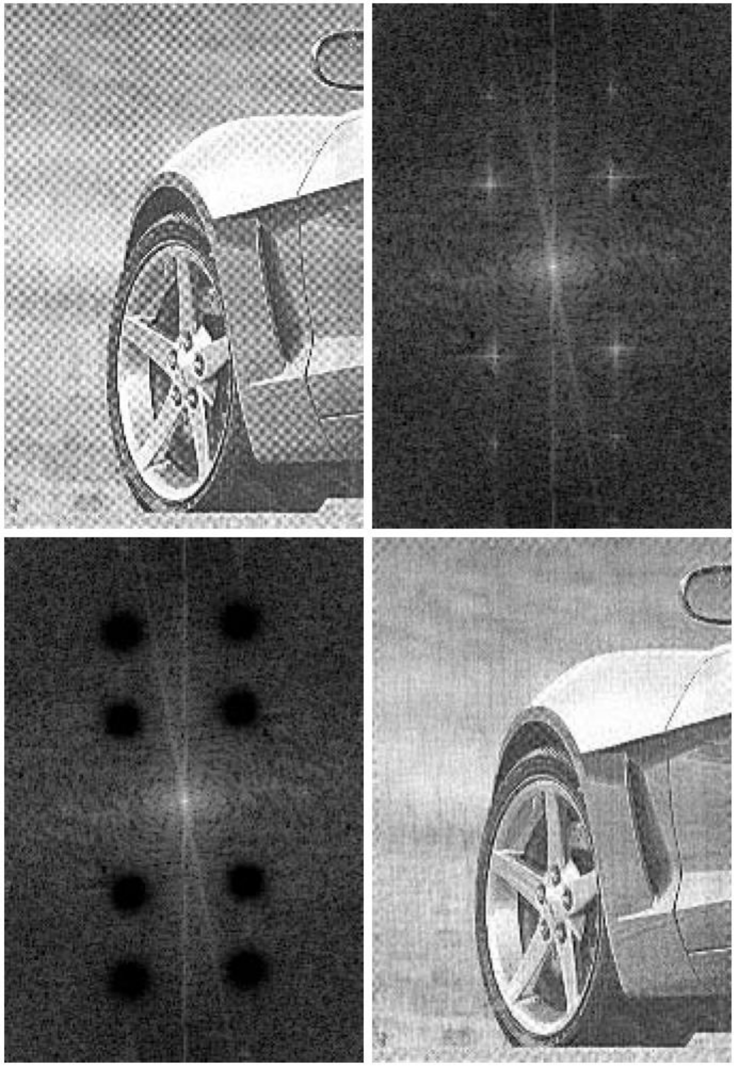
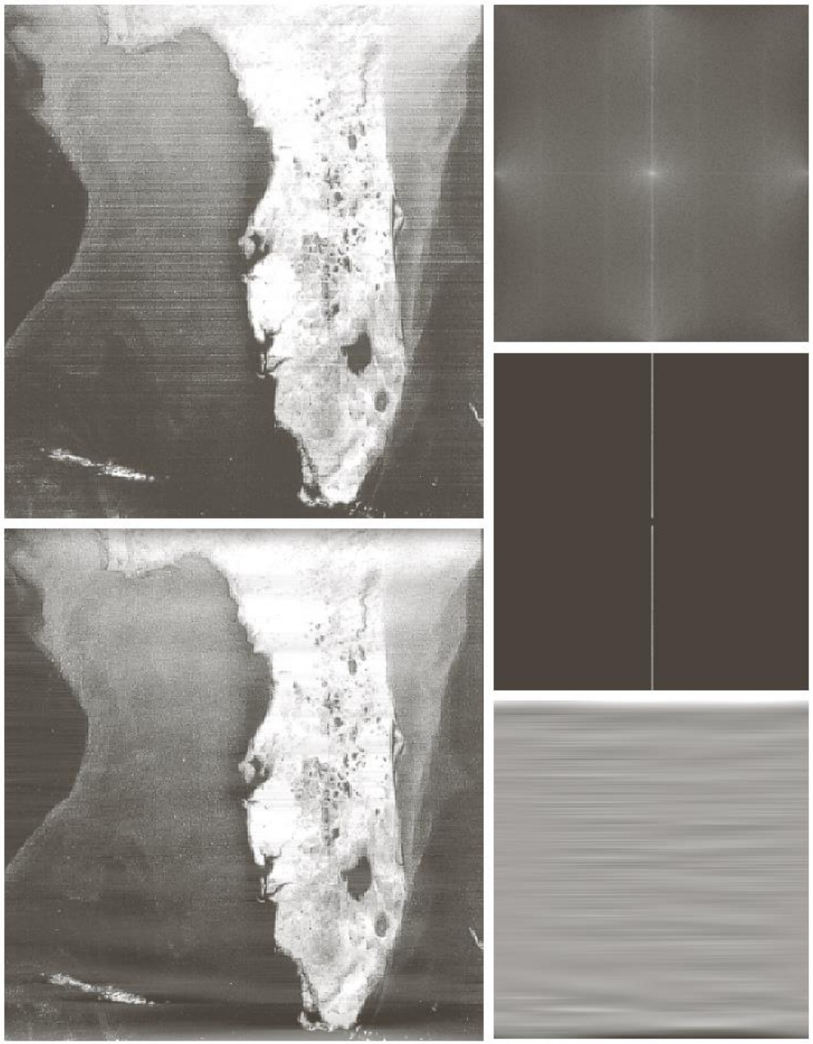


FIGURE 4.64 (a) Sampled newspaper image showing a moiré pattern. (b) Spectrum. (c) Butterworth notch reject filter multiplied by the Fourier transform. (d) Filtered image.

Example of notch filter



a b c e d

FIGURE 5.19 (a) Satellite image of Florida and the Gulf of Mexico showing horizontal scan lines. (b) Spectrum. (c) Notch pass filter superimposed on (b). (d) Spatial noise pattern. (e) Result of notch reject filtering. (Original image courtesy of NOAA.)

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  - Wiener filters
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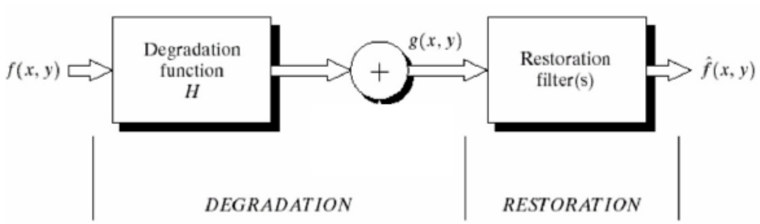
Recover from linear degradation

- Degradation function
  - Linear (eq 5.5-3, 5.5-4)
    - Homogeneity
    - Additivity
  - Position-invariant (in cartesian coordinates, eq 5.5-5)

→ linear filtering with H(u,v)
convolution with h(x,y) – point spread function



$$g(x,y) = H[f(x,y)] + \underbrace{\eta(x,y)}_{=\,0}$$

Divide-and-conquer step #2: linear degradation, noise negligible.

Estimate the degradation Function

- By Image Observation
- By Experimentation
- By modeling

Example of turbulence model

a b
c d

FIGURE 5.25 Illustration of the atmospheric turbulence model. (a) Negligible turbulence. (b) Severe turbulence, $k = 0.0025$. (c) Mild turbulence, $k = 0.001$. (d) Low turbulence, $k = 0.00025$. (Original image courtesy of NASA.)

Image blur due to motion

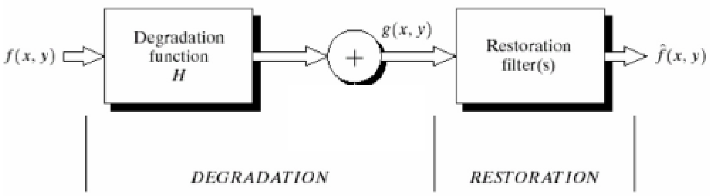
a b

FIGURE 5.26 (a) Original image. (b) Result of blurring using the function in Eq. (5.6-11) with $a = b = 0.1$ and $T = 1$.

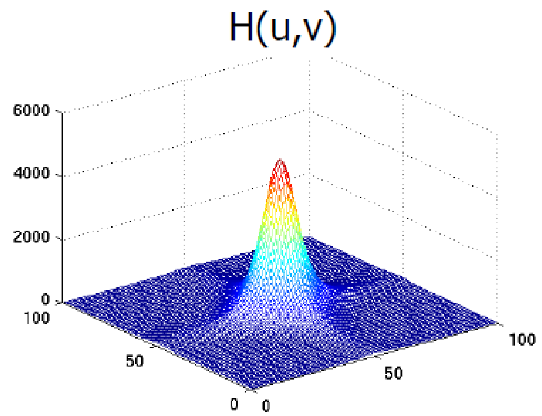
G&W Equ. 5.6-11

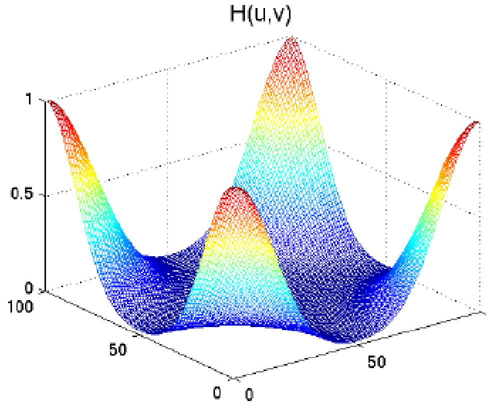
Image Restoration by Inverse filter

- assume h is known: low-pass filter H(u,v)



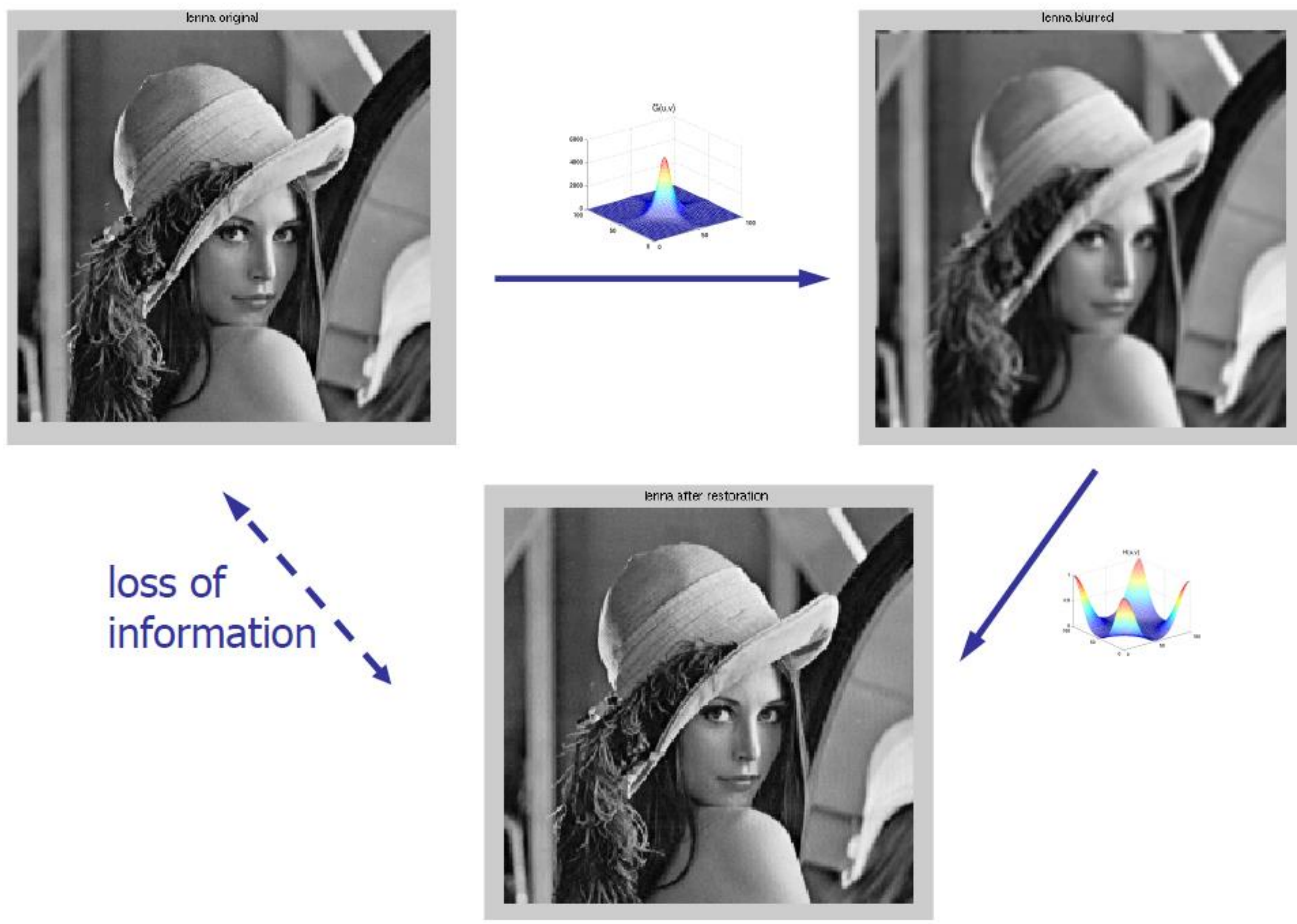
- inverse filter $\hat{H}(u,v) = 1/H(u,v)$
- recovered image $\hat{F}(u,v) = G(u,v)\hat{H}(u,v)$



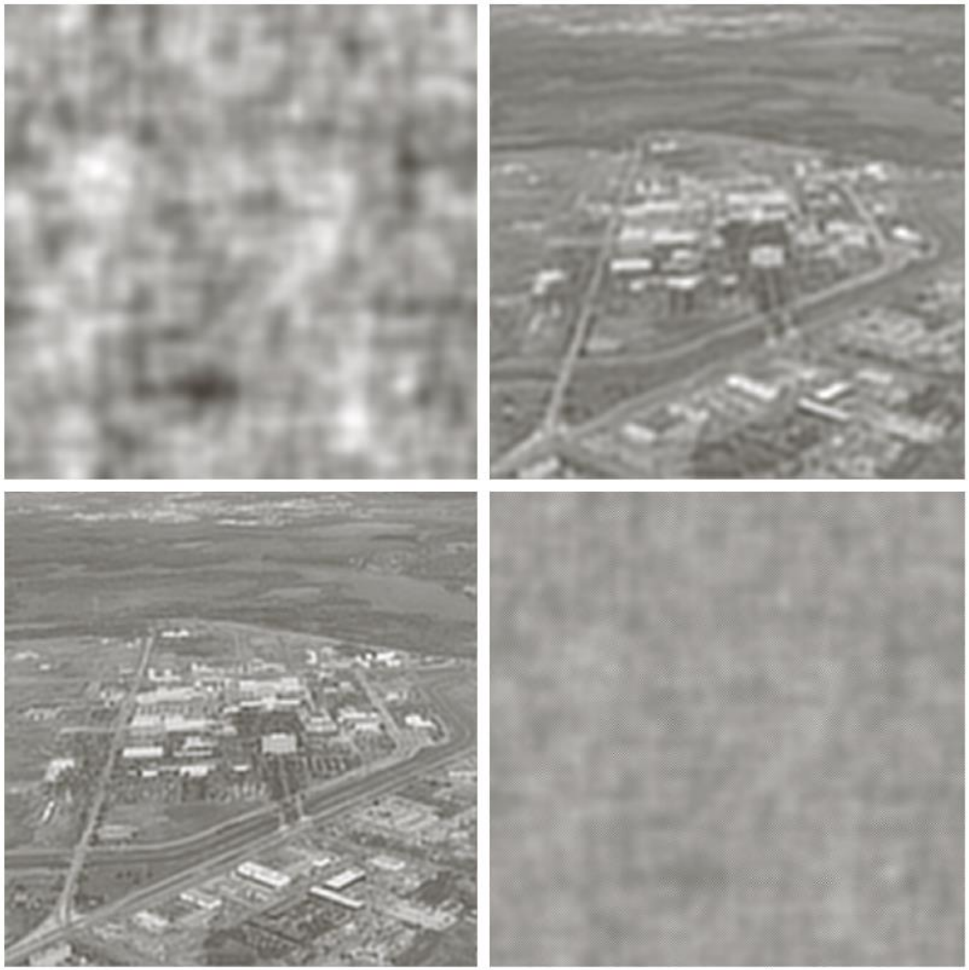


[EE381K, UTexas]

Inverse filtering example



Inverse filtering example



a b
c d

FIGURE 5.27 Restoring Fig. 5.25(b) with Eq. (5.7-1). (a) Result of using the full filter. (b) Result with H cut off outside a radius of 40; (c) outside a radius of 70; and (d) outside a radius of 85.

Input image

remedy 1:
inverse filter with cut-off

$$\hat{H}(u,v) = \begin{cases} 1/H(u,v), & |D(u,v)| \leq \varepsilon \\ 0, & |D(u,v)| > \varepsilon \end{cases}$$

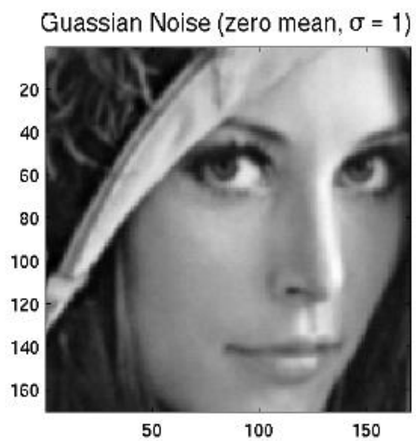
Inverse filtering under noise

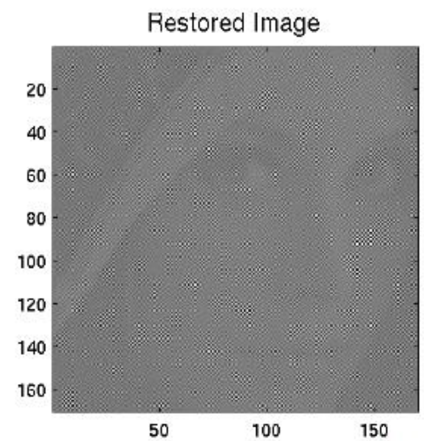
- in reality, we often have
  - H(u,v) = 0, for some u, v. e.g. motion blur
  - noise N(u,v) ≠ 0

$$\hat{H}(u,v) = 1/H(u,v)$$
$$\hat{F}(u,v) = G(u,v)\hat{H}(u,v)$$

$\Rightarrow$

$$\begin{aligned} G(u,v) &= F(u,v)H(u,v) + N(u,v) \\ \hat{F}(u,v) &= F(u,v) + \frac{N(u,v)}{\hat{H}(u,v)} \end{aligned}$$



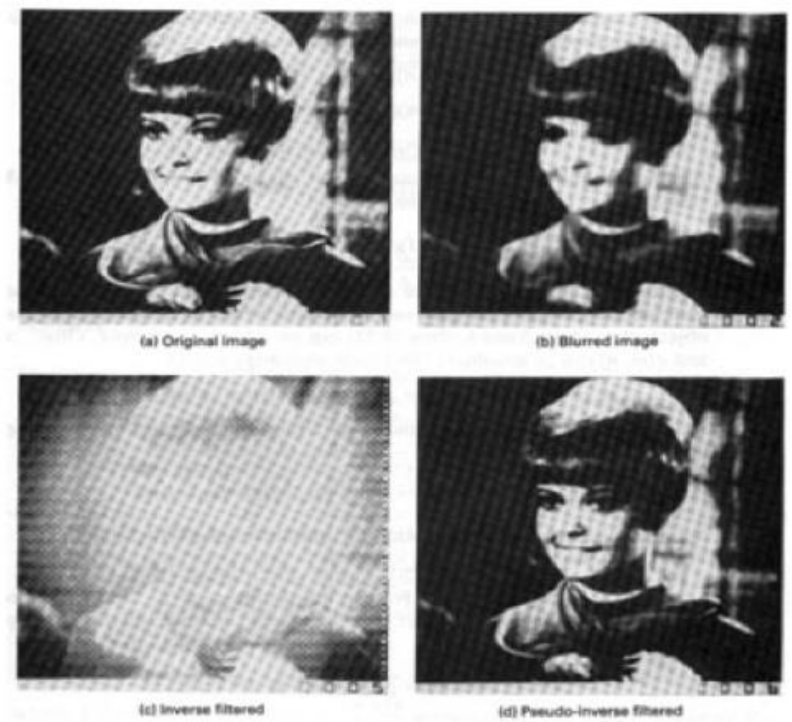


[EE381K, UTexas]

Pseudo-inverse filtering

cut-off based on fiter frequency

$$\hat{H}(u,v) = \begin{cases} 1/H(u,v), & |H(u,v)| \geq \epsilon \\ 0, & |H(u,v)| < \epsilon \end{cases}$$



Back to the original problem

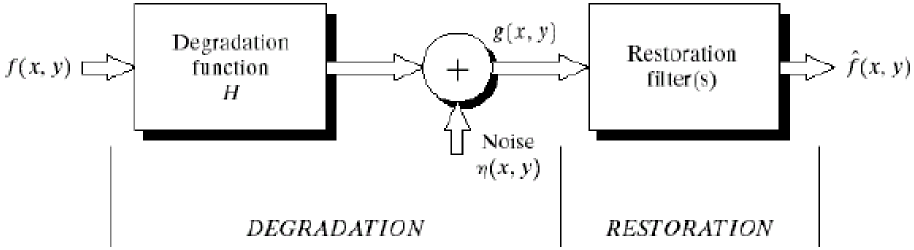


FIGURE 5.1 A model of the image degradation/ restoration process.

Inverse filter with cut-off:

$$\hat{H}(u,v) = \begin{cases} 1/H(u,v), & |D(u,v)| \le \varepsilon \\ 0, & |D(u,v)| > \varepsilon \end{cases}$$

Pseudo-inverse filter:

$$\hat{H}(u,v) = \begin{cases} 1/H(u,v), & |H(u,v)| \ge \epsilon \\ 0, & |H(u,v)| < \epsilon \end{cases}$$

- Can the filter take values between 1/H(u,v) and zero?
- Can we model noise directly?

Wiener (minimum mean square error) filter

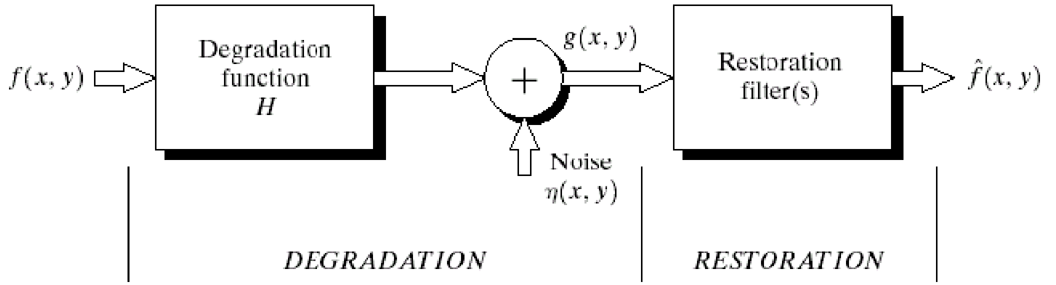


FIGURE 5.1 A model of the image degradation/restoration process.

$W(u, v)$

- goal: restoration with minimum mean-square error (MSE)

$$\min_{W} \quad e^2 = E\{(f - \hat{f})^2\}$$

- optimal solution (nonlinear):

$$\hat{f}(x, y) = E\{f(x, y) | g(m, n), \forall(m, n)\}$$

- restrict to linear space-invariant filter

$$\hat{f}(x, y) = w(x, y) * g(x, y)$$

- find "optimal" linear filter W(u,v) with min. MSE …

Wiener filter

- goal: restoration with minimum mean-square error (MSE)

$$\min_{W} \quad e^2 = E\{(f - \hat{f})^2\}$$

$$\hat{f}(x,y) = w(x,y) * g(x,y)$$

- find "optimal" linear filter W(u,v) with min. MSE

→ orthogonal condition $\quad E\{g(f - \hat{f})\} = 0$

→ correlation function $\quad R_{fg}(x,y) = w(x,y) * R_{gg}(x,y)$

→ wide-sense-stationary (WSS) signals

$$R_{fg}(x_1, y_1, x_2, y_2) = E\{f(x_1, y_1) g(x_2, y_2)\} \xrightarrow{WSS} R_{fg}(x_1 - x_2, y_1 - y_2)$$

→ Fourier Transform: from correlation to spectrum

$$S_{fg}(u,v) = \mathcal{F}\{R_{fg}(x,y)\}, \;\; S_{gg}(u,v) = \mathcal{F}\{R_{gg}(x,y)\}$$

⇨ $$W(u,v) = \frac{S_{fg}(u,v)}{S_{gg}(u,v)} = \frac{H^*(u,v) S_{ff}(u,v)}{|H(u,v)|^2 S_{ff}(u,v) + S_{\eta\eta}(u,v)}$$

S_{ff} and $S_{\eta\eta}$ are the power spectra of the signal and noise, respectively

SNR and MSE measurements

Signal-to-noise ratio:

$$SNR = \frac{\sum_{u=0}^{M-1} \sum_{v=0}^{N-1} |F(u,v)|^2}{\sum_{u=0}^{M-1} \sum_{v=0}^{N-1} |N(u,v)|^2}$$

$$SNR = \frac{\sum_{u=0}^{M-1} \sum_{v=0}^{N-1} \hat{f}(x,y)^2}{\sum_{u=0}^{M-1} \sum_{v=0}^{N-1} [f(x,y) - \hat{f}(x,y)]^2}$$

Mean square error:

$$MSE = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} [f(x,y) - \hat{f}(x,y)]^2$$

Observations about Wiener filter

$$
\begin{aligned}
W(u,v) &= \frac{H^*(u,v)S_{ff}(u,v)}{|H(u,v)|^2 S_{ff}(u,v) + S_{\eta\eta}(u,v)} \\
&= \frac{1}{H(u,v) + \frac{S_{\eta\eta}}{H^*(u,v)S_{ff}}}
\end{aligned}
$$

- If no noise, $S_{\eta\eta} \rightarrow 0$ $\quad W(u,v)|_{S_{\eta\eta}\rightarrow 0} = \begin{cases} \frac{1}{H(u,v)}, & H(u,v) \neq 0 \\ 0, & H(u,v) = 0 \end{cases}$

  → Pseudo inverse filter

- If no blur, H(u,v)=1 (Wiener smoothing filter)

$$
W(u,v)|_{H=1} = \frac{1}{1 + S_{\eta\eta}(u,v)/S_{ff}(u,v)} = \frac{SNR(u,v)}{SNR(u,v)+1}
$$

  → More suppression on noisier frequency bands

1-D Wiener Filter Shape

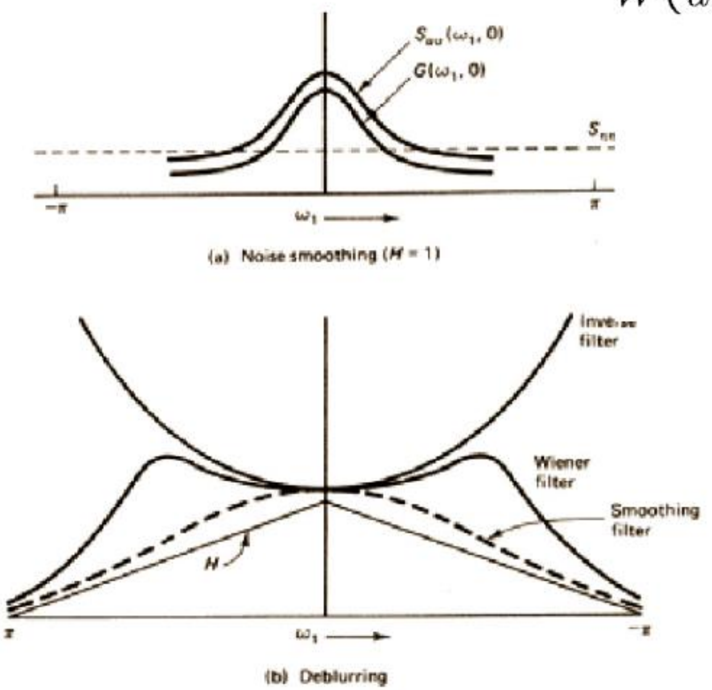


Figure 8.11 Wiener filter characteristics.

Wiener Filter implementation

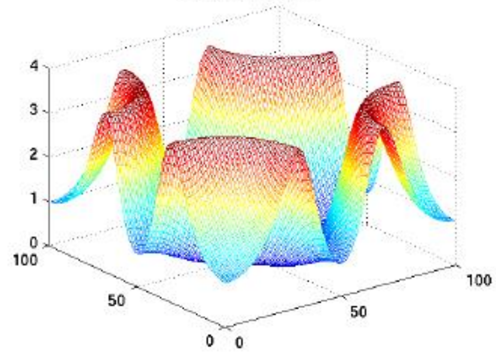
$$
\begin{aligned}
W(u,v) &= \frac{H^*(u,v)S_{ff}(u,v)}{|H(u,v)|^2 S_{ff}(u,v) + S_{\eta\eta}(u,v)} \\
&= \frac{H^*(u,v)}{|H(u,v)|^2 + \frac{S_{\eta\eta}}{S_{ff}}} \\
&= \frac{H^*(u,v)}{|H(u,v)|^2 + K}
\end{aligned}
$$

Where K is a constant (w.r.t. u and v) chosen according to our knowledge of the noise level.

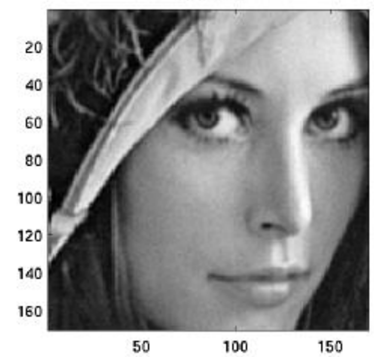
Wiener Filter Examples

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K}$$

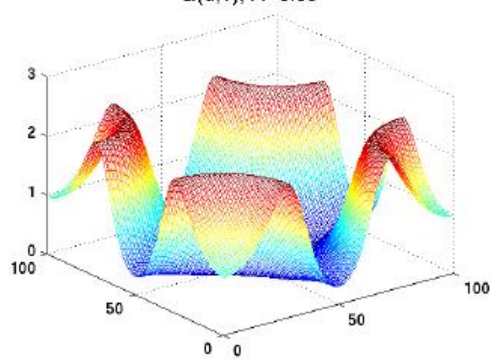
G(u,v), K=0.02



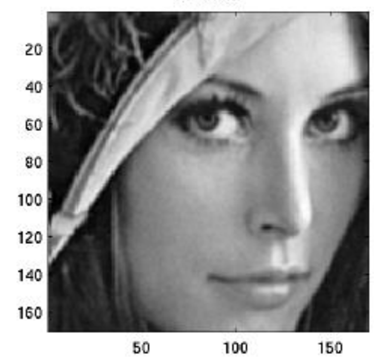
K=0.02



G(u,v), K=0.05



K=0.05



Wiener filter examples

a b c

FIGURE 5.28 Comparison of inverse and Wiener filtering. (a) Result of full inverse filtering of Fig. 5.25(b). (b) Radially limited inverse filter result. (c) Wiener filter result.

- Wiener filter is more robust to noise, and preserves high-frequency details.

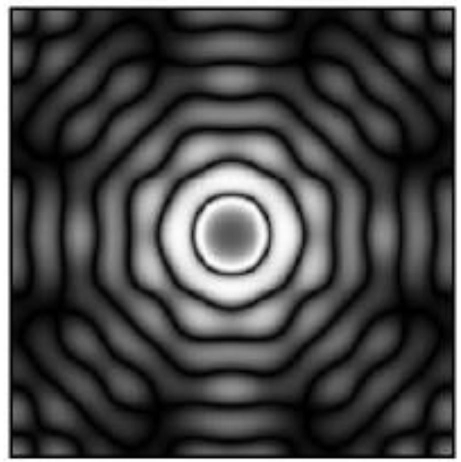
Wiener filter examples

(a) Blurry image (b) restored w. regularized pseudo inverse
(c) restored with wiener filter

Wiener filter: when does it not work

How much de-blurring is just enough?

image 'blurr1'



wiener filter

restored license plate

Improve Wiener filters

- geometric mean filters

$$W(u,v) = \left[\frac{H*(u,v)}{|H(u,v)^2|}\right]^{\alpha} \left[\frac{H^*(u,v)}{|H(u,v)|^2 + \beta\frac{S_{\eta\eta}(u,v)}{S_{ff}(u,v)}}\right]^{1-\alpha}$$

- Constrained Least Squares
  - Wiener filter emphasizes high-frequency components, while images tend to be smooth

$$\min_{f} \; |g - H\hat{f}|^2 + \alpha|C\hat{f}|^2$$

$\hat{f}$: the estimate for undegraded image

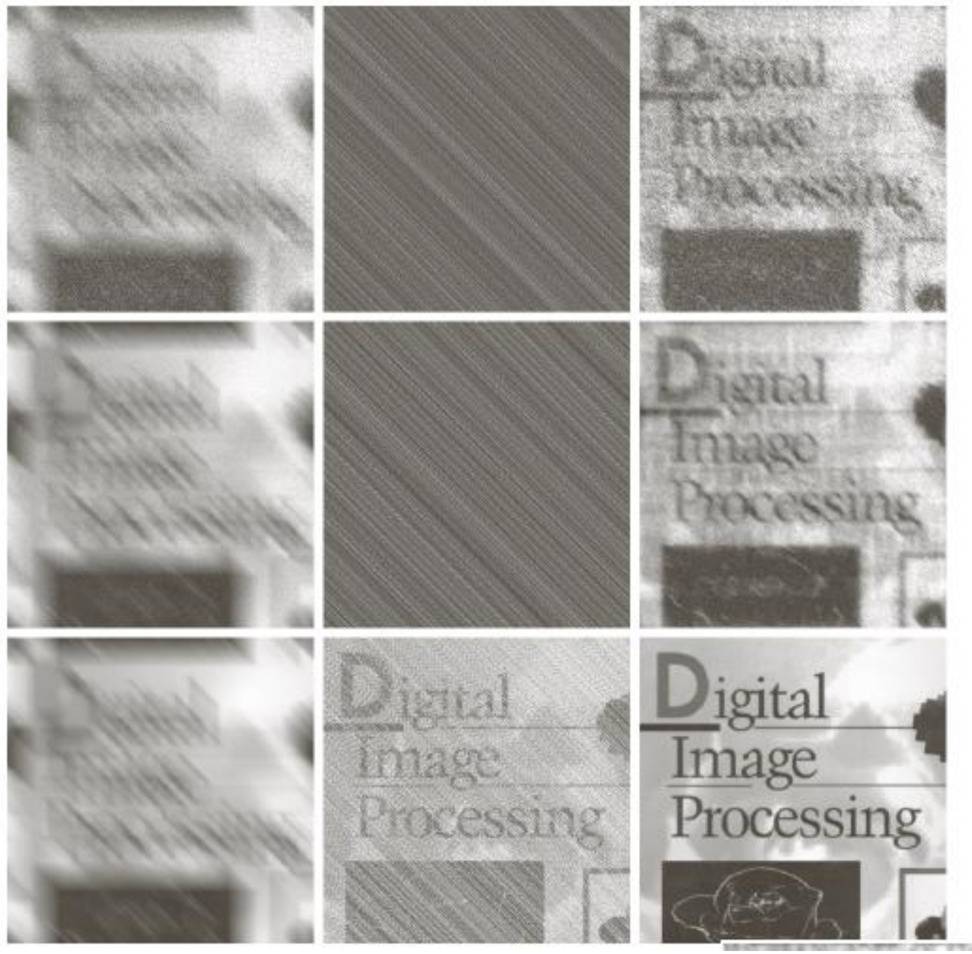
$C\hat{f}$: a high-passed version of $\hat{f}$

degraded inverse-filtered Wiener-filtere

motion blur
+ noise

noise*10^{-1}

noise*10^{-5}



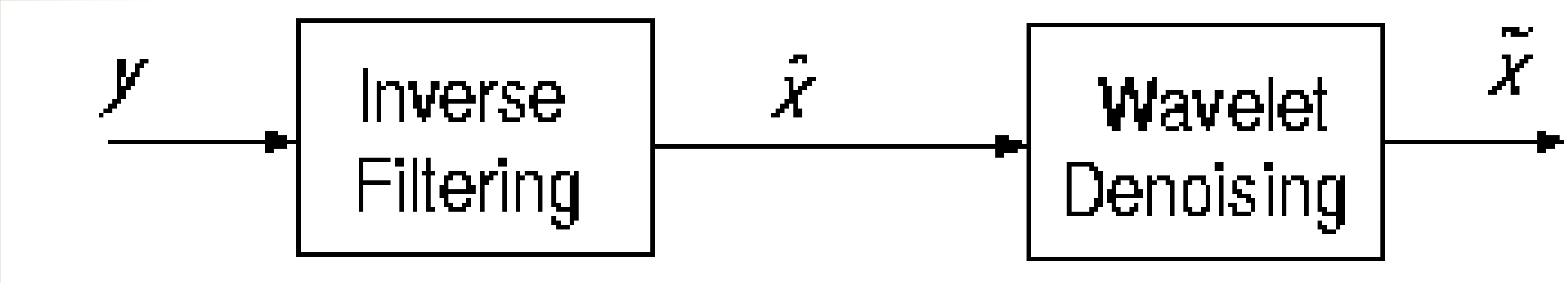
Example of improved wiener filter

a b c

FIGURE 5.30 Results of constrained least squares filtering. Compare (a), (b), and (c) with the Wiener filtering results in Figs. 5.29(c), (f), and (i), respectively.

Wavelet Restoration

- Two steps:
  - Fourier-domain inverse filtering
  - Wavelet-domain image denoising.
- When the blurring function is not invertible, the algorithm is not applicable

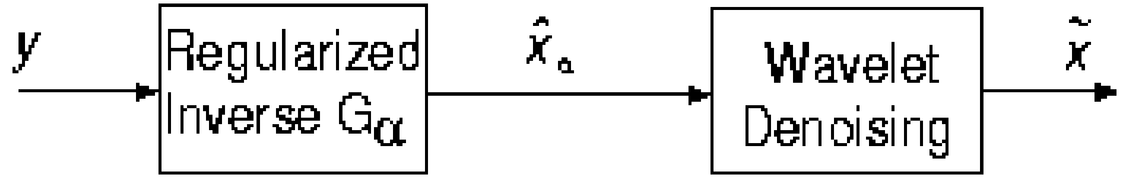


Wavelet-based deconvolution technique for ill-conditioned systems

- Two steps:
  - modifying the Wiener filter

$$G_\alpha = \frac{H^* S_{xx}}{|H|^2 S_{xx} + \alpha S_{\eta\eta}}.$$

  - Wavelet-domain image denoising.



Example of Wavelet Restoration

Blurred Lena Image PSNR = 23.2993, MSE = 304.1938

Restored lena Image (Wavelet) PSNR = 16.8552, MSE = 1341.4

Restored Lena Image (WaRD) PSNR = 19.5115, MSE = 727.7

Restored Lena Image (Subband) PSNR = 20.1223, MSE = 632.2

Power Spectrum Equalization (PSE)

- We assumed that we knew the blurring function **h**, what should we do if they were unknown?
  - Wiener Filter

$$G = \frac{H^* S_{uu}}{|H|^2 S_{uu} + S_{\eta\eta}}$$

  - Power Spectrum Equalization -- restore the power spectrum of the degraded image

$$G = \left[\frac{S_{uu}}{|H|^2 S_{uu} + S_{\eta\eta}}\right]^{1/2}$$

Example of PSE

Lenna after Blurring Plus Noise, Mean Squared Error = 1.0660e+05

Lenna restored using Wiener Filter, Mean Squared Error = 123.2

Lenna restored using PSE, Mean Squared Error = 419.5

Blind deconvolution

- Restore image without knowledge of our blurring function
- Estimate H (Jain approach)

$$\log|H| = \frac{1}{M}\sum_{k=1}^{M}[\log|V_k| - \log|U_k|]$$

- Wiener filter

http://www.owlnet.rice.edu/~elec539/Projects99/BACH/proj2/blind/bd.html

Example of Blind deconvolution

Lenna after Blurring Plus Noise
Mean Squared Error = 1.0660e+05

Lenna restored using Jain Approach
Mean Squared Error = 283.1

Outline

- What is image restoration
  - Scope, history and applications
  - A model for (linear) image degradation
- Restoration from noise
  - Different types of noise
  - Examples of restoration operations
- Restoration from linear degradation
  - Inverse and pseudo-inverse filtering
  - Wiener filters
  - Wavelet Restoration
  - Blind de-convolution
- Geometric distortion and its corrections

Geometric distortions

- Modify the spatial relationships between pixels in an image
- a. k. a. "rubber-sheet" transformations

- Two basic steps
  - Spatial transformation
  - Gray-level interpolation

$$x' = r(x, y)$$
$$y' = s(x, y)$$

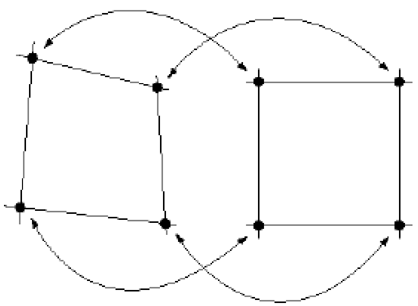


FIGURE 5.32 Corresponding tiepoints in two image segments.

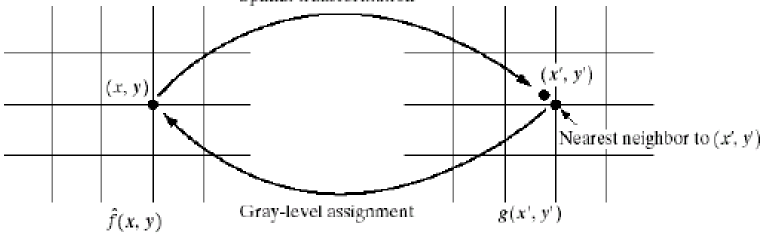
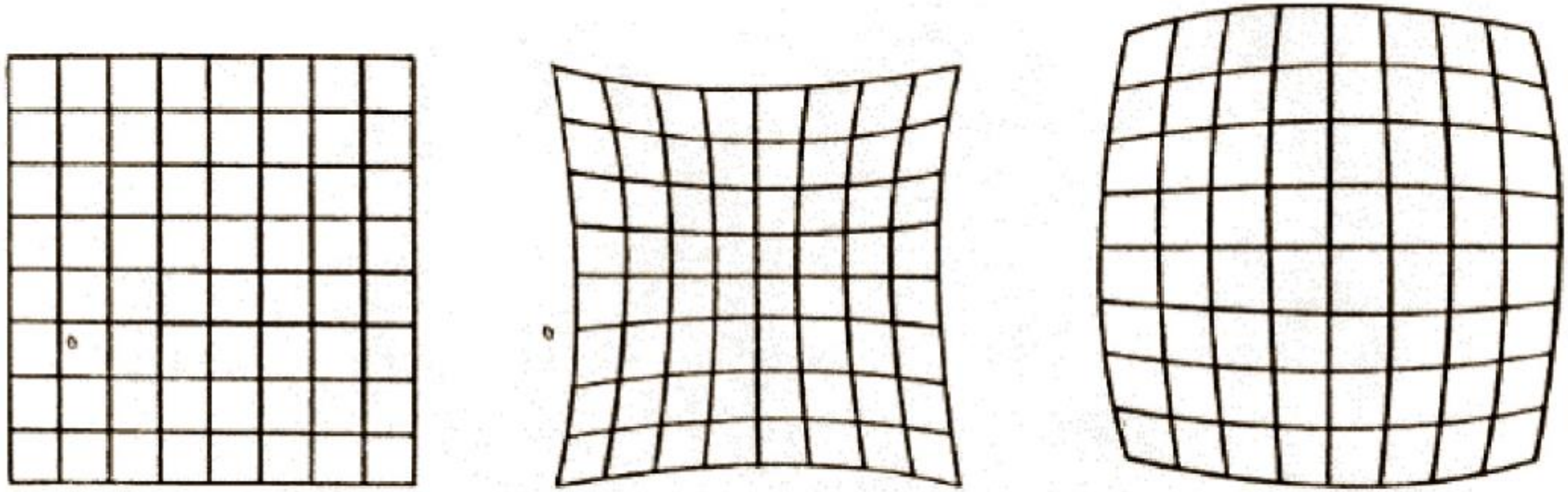


FIGURE 5.33 Gray-level interpolation based on the nearest neighbor concept.

Geometric/spatial distortion examples



(a) Original (b) Pincushion distortion (c) Barrel distortion

FIGURE 14.2-1. Example of geometric distortion.

Recovery from geometric distortion

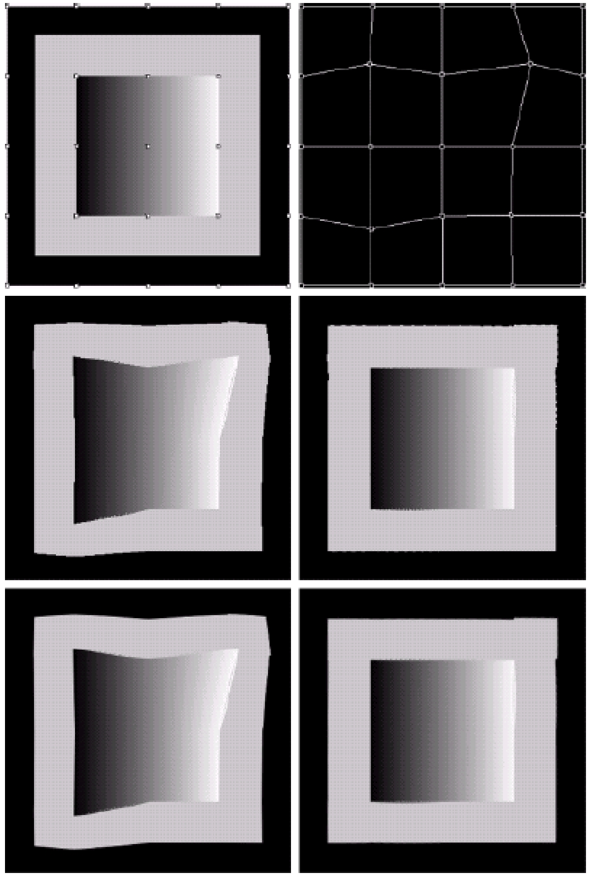


FIGURE 5.34 (a) Image showing tiepoints. (b) Tiepoints after geometric distortion. (c) Geometrically distorted image, using nearest neighbor interpolation. (d) Restored result. (e) Image distorted using bilinear interpolation. (f) Restored image.

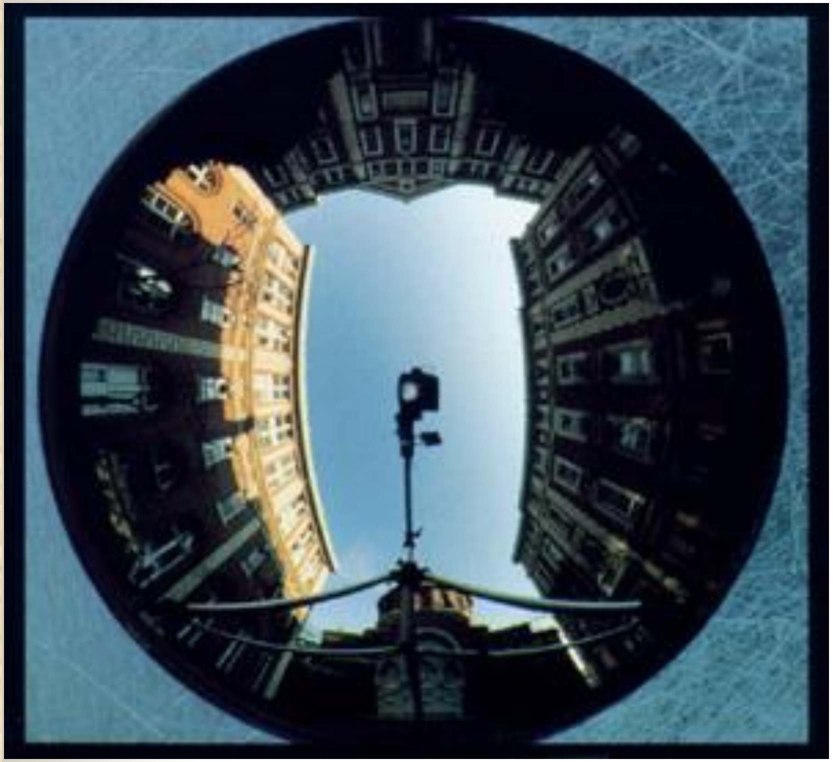
Recovery from geometric distortion

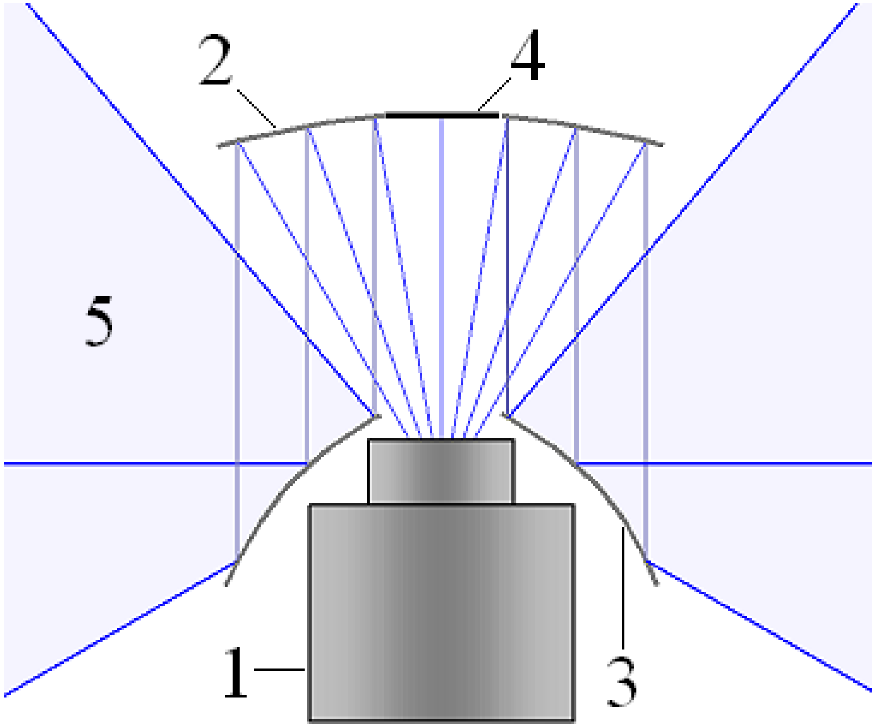
(a) (b)

Fig. 5. (c) Image produced by a Computar 2.5mm lens and a Computar $1/3''$ CCD board camera. (b) Distortion parameters recovered via the minimization of ξ_3 are used to map (a) to perspective image. Notice that straight lines in the scene, such as door edges, map to straight lines in the undistorted images.

Rahul Swaminathan, Shree K. Nayar: Nonmetric Calibration of Wide-Angle Lenses and Polycameras. IEEE Trans. Pattern Anal. Mach. Intell. 22(10): 1172-1178 (2000)

Omnicamera





1. Camera 2. Upper Mirror 3. Lower Mirror
4. "Black Spot" 5. Field of View (light blue)

https://en.wikipedia.org/wiki/Omnidirectional_(360-degree)_camera

Estimating distortions

- calibrate
- use flat/edge areas
- … ongoing work

a. Original
BlurExtent = 0.0104

b. Out-of-focus
BlurExtent = 0.4015

c. Original
BlurExtent = 0.0462

d. Linear-motion
BlurExtent = 0.2095

http://photo.net/learn/dark_noise/

[Tong et. al. ICME2004]

Summary

- Image degradation model
- Restoration from noise
- Restoration from linear degradation
- Inverse and pseudo-inverse filters, Wiener filter, constrained least squares, wavelet restoration,
- Geometric distortions
- Readings
  - G&W Chapter 5.1 – 5.10.
  - M. R. Banham and A. K. Katsaggelos "Digital Image Restoration“, IEEE Signal Processing Magazine, vol. 14, no. 2, Mar. 1997, pp.24-41

Who said distortion is a bad thing?

blur ...

noise ...

geometric ...

Suggestions for Final Project -- 1

- Each student or team works on different projects
- **Send me the TITLE and a brief description if you will choose the final project by yourself before 3/20/2026 by email (**must be image processing related**).**
- No more than two students work on same project
  - Teamwork
  - Need to show the contributions for each student

Suggestions for Final Project -- 2

- Your final project will be started on 3/25 after HW3 due.
- For the project:
  - Code
  - Report (I'll send out the template)
  - Presentation

Announcement:

- Ali will present HW2 on **3/4.**
- HW3 is out today, due on 3/24.
- **Midterm Exam: Mar. 11, 2026**.

HW 2 Presentation